

CeramTec ZN 40 Zirconia, ZrO2-MgO

Categories: [Ceramic](#); [Oxide](#); [Zirconium Oxide](#)

Material Notes: ZN 40 is a zirconium oxide ceramic with an excellent surface finish and a higher strength than aluminum oxide. Its good crack resistance and anti-friction properties, combined with steel-like thermal expansion attributes, make this material an ideal partner for bearing composites working at elevated temperatures.

Key Words: Zirconium Oxide

Vendors: No vendors are listed for this material. Please [click here](#) if you are a supplier and would like information on how to add your listing to this material.

Physical Properties	Metric	English	Comments
Density	5.74 g/cc	0.207 lb/in ³	DIN EN 623-2
Water Absorption	0.00 %	0.00 %	Open Porosity; DIN EN 623-2
Permeability	0.00	0.00	%, Gas
Weibull Modulus	25	25	DINV ENV 843-5

Mechanical Properties	Metric	English	Comments
Vickers Microhardness	1240	1240	HV1; DINV ENV 843-4
Tensile Strength at Break	500 MPa	72500 psi	ACMA Test #4 / DIN EN 843-1
Tensile Modulus	210 GPa	30500 ksi	Young's; DINV ENV 843-2
Flexural Strength	500 MPa	72500 psi	DIN EN 843-1
Compressive Strength	1600 MPa	232000 psi	DIN 51067T1
Poissons Ratio	0.30	0.30	DINV ENV 843-2
Fracture Toughness	8.10 MPa-m ^{1/2}	7.37 ksi-in ^{1/2}	K _{IC} (SEVNB); DIN CEN/TS 14425-1
Shear Modulus	80.8 GPa	11700 ksi	Calculated

Electrical Properties	Metric	English	Comments
Volume Resistivity 	500000 ohm-cm @Temperature 400 °C	500000 ohm-cm @Temperature 752 °F	IEC 672-1
	5.00e+13 ohm-cm @Temperature 20.0 °C	5.00e+13 ohm-cm @Temperature 68.0 °F	IEC 672-1
Dielectric Constant	27 @Frequency 1e+6 Hz	27 @Frequency 1e+6 Hz	IEC 672-1
Dielectric Strength	19.0 kV/mm	483 kV/in	IEC 672-1
Dielectric Loss Index	0.020 @Frequency 9.00e+9 Hz	0.020 @Frequency 9.00e+9 Hz	IEC 672-1

Thermal Properties	Metric	English	Comments
CTE, linear 	10.2 µm/m-°C @Temperature 20.0 - 400 °C	5.67 µin/in-°F @Temperature 68.0 - 752 °F	DIN EN 821-1
	11.0 µm/m-°C @Temperature 20.0 - 1000 °C	6.11 µin/in-°F @Temperature 68.0 - 1830 °F	DIN EN 821-1
Specific Heat Capacity	0.400 J/g-°C	0.0956 BTU/lb-°F	DINV ENV 821-3
Thermal Conductivity	3.00 W/m-K	20.8 BTU-in/hr-ft ² -°F	DIN EN 821-2

Maximum Service Temperature, Air	850 °C	1560 °F
Maximum Service Temperature, Inert	850 °C	1560 °F

Descriptive Properties

Ra = Arithmetic Mean Roughness Value (µm)	<0.06	
Thermal Shock Resistance R1 (K)	163	calculated; $R1 = [\sigma_s (1-\mu)] / (\alpha \cdot E)$

Some of the values displayed above may have been converted from their original units and/or rounded in order to display the information in a consistent format. Users requiring more precise data for scientific or engineering calculations can click on the property value to see the original value as well as raw conversions to equivalent units. We advise that you only use the original value or one of its raw conversions in your calculations to minimize rounding error. We also ask that you refer to MatWeb's [terms of use](#) regarding this information. [Click here](#) to view all the property values for this datasheet as they were originally entered into MatWeb.